# MGMTMSA 400 MIDTERM NOTES - FRONT

## CHAPTER 1: INTRODUCTION

- Q4. A startup ran an A/B test to compare two email subject lines. The experimental emails were sent to a test sample of their 200 most active users, with 50% randomly assigned to each email. Subject Line A led to a 15% higher click-through rate than Subject Line B, with a p-value of 0.048 from a two-sample t-test. They concluded they should send Subject Line A to the entire customer base. What is the biggest issue with the study design? Steps: inside the test, the random assignment is fine, there IS a comparison group (the B email), and the test itself fits the data. The flaw is who got sampled: the 200 most active users are not a random sample of the customer base, so the result may not generalize to everyone. Answer: A, the sample is not random, which limits generalizability.
- Q5. Which of the following analyses is supervised learning? (A) Walmart groups customers into segments based on purchase patterns. (B) Outlook classifies whether a new email is spam or not spam based on the email's text. (C) Wayfair identifies the key features of its furniture products using image pixels. (D) Spotify discovers groups of songs with similar acoustic features. Steps: supervised learning needs a NAMED outcome present in the training data. Outlook's spam labels are exactly that: past labeled emails teach it to classify new ones. Segments, discovered groups, and extracted features have no outcome variable, so they are unsupervised. Answer: B.
- Q15. Which of the following would pose the most serious threat to establishing causality from the Advantage+ regression? (A) In 2022 and 2023 the advertiser used two different ad agencies. (B) Sales often spike during Thanksgiving and Christmas. (C) The advertiser's customers are a niche market. (D) The advertiser randomly runs promotions across 2024. (E) The advertiser plans to launch a new product next year. Steps: Advantage+ turned on July 1, 2024, so the entire 'after' period includes the holiday season while the 'before' period does not. Holiday spikes are a time confounder pushing both the treatment timing and sales, which can fake an Advantage+ effect. Agencies from earlier years and next year's product cannot affect 2024 data, a niche market limits generalizability rather than causality, and random promotions are spread evenly. Answer: B.

## CHAPTER 2: REVIEW OF STATISTICS

- Q1. Suppose there is a random variable X ~ N(1, 4). What is the standard deviation of X? Steps: in the notation N(a, b), the first slot a is the MEAN and the second slot b is the VARIANCE, not the SD. Here the variance is 4, so the SD is its square root: sqrt(4) = 2. Traps: 1 is the mean and 4 is the variance; neither one is the SD. Answer: B (2).
- Q2. BrightMart, a large online retailer, introduced a personalized recommendation engine to increase customer spending. Historically, the average order value (AOV) per transaction has been $50. After implementing the system, the company collects a random sample of 1,000 transactions from the past month and finds the new AOV is slightly higher. The team wants to determine whether this represents a meaningful increase in AOV. What is the best statistical analysis? Steps: there is ONE new sample and ONE known benchmark value ($50), so it is a one-sample test on a mean. The question is directional, an increase only, so one-tailed. Sample mean with unknown SD means a t-test. Answer: A, one-sample, one-tailed t-test.
- Q3. A credit card company is designing a card to appeal to young urban customers who do not currently have credit cards. They want to know how much this target population typically spends per month. They plan a survey and want to ensure the 95% confidence interval is not wider than $1,000. From similar customers, the SD of spending is $2,000. To how many people should they send the survey? (The exam gives z0.975 = 1.96.) Steps: the interval width is two margins, so the margin is 1000 / 2 = 500. Solve margin = z x SD / sqrt(n) backwards: n >= (1.96 x 2000 / 500)^2 = (7.84)^2 = 61.47. Always round UP. Answer: B (62).
- Q6. Suppose X ~ N(1, 3) and Y ~ N(1, 1), and the correlation between X and Y is 0.6. What is the expected value (population mean) of X - 2Y? Steps: expected values pass through any linear combination using the same weights; correlation is irrelevant for a mean. E[X - 2Y] = E[X] - 2 E[Y] = 1 - 2(1) = -1. Since -1 is not one of the listed numeric choices (1, 2.6, 3, not enough info), the answer is 'none of the above.' Answer: E.
- Q7. Continue with X ~ N(1, 3), Y ~ N(1, 1), corr(X, Y) = 0.6. What is the variance of X - 2Y? Steps: first get the covariance: Cov = corr x SDx x SDy = 0.6 x sqrt(3) x 1 = 1.04. Then use Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2ab Cov, with a = 1 and b = -2: Var = 1(3) + 4(1) + 2(1)(-2)(1.04) = 3 + 4 - 4.16 = 2.84. The choices are 5, 7.16, 7.6, and not enough info; 2.84 is not there. Answer: E, none of the above.
- Q8. You are estimating average household income in Los Angeles County from random samples. Which statements are true? (Select all that apply.) (A) Compared to the normal distribution, household income in the population likely has a heavy tail on the right. (B) The sample mean of household incomes will be approximately normal when the sample size is large. (C) The CLT guarantees individual incomes in the sample will be normally distributed for large samples. (D) As n increases, the sample mean gets closer to the true MEDIAN income. Steps: incomes are right-skewed, so (A) is true. The CLT is about the distribution of the sample MEAN, making (B) true and (C) false: the CLT never makes the raw data normal. And means settle at the population mean, not the median, so (D) is false. Answer: A and B.
- Q9. Jane runs an all-you-can-eat restaurant charging $19 per customer. From thousands of records, her cost per customer follows a normal distribution with mean $17.50 and standard deviation $3.00. What is the best point predictor of the cost for the next customer? Steps: for a single future draw from a Normal distribution, the best point prediction is the center of that distribution, its mean. The SD (3.00) measures spread around the prediction, not the prediction itself. Answer: B (17.50).
- Q10. Jane wants a 90% prediction interval of customer costs, centered around the mean ($17.50), so that 90% of customers' costs fall within this range. What is the width of that interval? Steps: this is an interval for ONE new customer's cost, so it uses the full SD of individual costs, not a standard error. A 90% interval uses the multiplier 1.64: interval = 17.50 +/- 1.64 x 3. Width = 2 x 1.64 x 3 = 9.84. Answer: D.
- Q11. Suppose customers are i.i.d. On one day Jane serves 100 customers, and on another day she serves 250 customers. Which day is more likely to have a higher average profit per customer (price minus cost per customer)? Steps: the expected average profit is the same on both days: $19 minus the $17.50 mean cost. The day with 250 customers has a SMALLER standard error, so its average sits closer to that common center (the Law of Large Numbers), but a tighter spread does not make the average higher on average. Each day is equally likely to land above the center. Answer: C, both days are equally likely.
- Q16. You are a pricing manager at Pepsi planning a survey in Sioux City, Iowa (about 90,000 residents) to estimate willingness to pay (WTP) for a 20-ounce Diet Pepsi. Your analyst makes claims about sample size and distributions. Which are correct? (Select all that apply.) (A) Because Sioux City residents may have similar preferences, a smaller sample may suffice for the same accuracy as in a more diverse city like New York. (B) A larger sample size generally allows more precise inference about average WTP. (C) No valid inference about average WTP is possible if WTP is not normally distributed. (D) No hypothesis test about the mean is possible unless WTP is exactly normal or t. Steps: similar preferences mean a smaller SD, and SE = SD / sqrt(n), so a smaller n reaches the same precision (A true). Bigger n always shrinks the SE (B true). The CLT makes sample means approximately normal for large n even when the raw WTP is not normal, so (C) and (D) are false. Answer: A and B.
- Q17. You plan to collect survey data such that all participants are independent and identically distributed (i.i.d.). Which of these practices does NOT violate i.i.d.? (A) Skipping participants who only drink Coke. (B) After surveying 10 white residents in a row, targeting an African American respondent so sample demographics match the city. (C) After finding a participant with a high WTP, looking for another with a low WTP to reduce sampling error. (D) Including respondents from neighboring towns outside Sioux City. Steps: (A) is selection on a trait, so not everyone in the population has the same chance to appear: not identical. (B) and (C) make the next draw depend on previous draws: not independent. (D) samples from a different population: not identical. Every listed practice breaks the assumption. Answer: E, all the practices above violate i.i.d.

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## CHAPTER 2 (CONTINUED)

- Q18. GLP-1 medications: Jonathan will start treatment only if the average 6-month effectiveness is at least 15% body-weight reduction. A pilot sample of 10 patients who completed 6 months on GLP-1 had body-weight losses (percentage points) of: 15.2, 14.1, 15.7, 15.4, 15.0, 17.2, 16.1, 14.9, 16.6, 14.8. What is the mean body-weight loss from the pilot sample? Steps: add all ten values: the sum is 155.0. Divide by the sample size: 155.0 / 10 = 15.5. Answer: C (15.5).
- Q19. Which of the following is the best null hypothesis for determining whether to prescribe GLP-1 to Jonathan? (A) The average 6-month body-weight reduction for GLP-1 users is equal to 15%. (B) The average 6-month body-weight reduction for GLP-1 users is lower than 15%. (C), (D), (E) The same claims about Jonathan's own predicted reduction. Steps: hypotheses are always about the POPULATION parameter, the average for GLP-1 users, never about one person's predicted outcome, so (C), (D), (E) are out. And because Jonathan prescribes only if the average is at least 15%, the claim that must be beaten, the null, is that the average is below that bar. Answer: B.
- Q20. You conduct a t-test for Jonathan and find the t-statistic is 1.71. What is your prescription decision at a 95% confidence level? (Provided t values: t9, 0.95 = 1.83 with df = 9.) Steps: the sample has n = 10, so df = 9. The test is one-sided (prescribe only if the average clears 15%), so the cutoff at the 5% level is 1.83. The observed t = 1.71 falls short of 1.83, so the data do not provide enough evidence that the average loss reaches 15%. Answer: C, there is not enough statistical evidence to support prescribing GLP-1 to Jonathan.

## CHAPTER 3: PREDICTIVE MODELS

- Q12. Advantage+ (Meta's AI ads product) regression of log_sales on the binary variable advantage: slope 0.08612 (t = 2.091), residual SE 0.7845 on 364 df, R^2 = 0.01186, mean of log_sales = 7.8415. The intercept (--a--) is blanked out. What is its estimated value, to two decimal places? Steps: the least squares line always passes through the point of sample means (x-bar, y-bar). So b0 = y-bar - b1 x-bar = 7.8415 - 0.08612 x (mean of advantage). Filling in the printed mean of advantage gives 7.84. Answer: C.
- Q14. The Advantage+ regression has a relatively small R^2 = 0.01186. Which of the following does that imply? (Select all that apply.) (A) The model is not valid. (B) Advantage explains only a small share of the variation in log_sales. (C) The advertiser should not use Advantage+. (D) The correlation between advantage and log_sales is not strong. (E) We should switch the dependent variable to unlogged sales. Steps: (B) is just the definition of R^2 restated, so it is implied. In a simple regression R^2 = r^2, so r is about sqrt(0.01186) = 0.11, a weak correlation, so (D) is implied. A small R^2 does not invalidate a model (A false), does not by itself make a business decision (C false, the slope was positive and significant), and says nothing about whether logging sales was right (E false). Answer: B and D.
- Q21. Election research: three separate simple regressions of Trump's county-level vote share (0 to 1), one per input. (1) Log average household income: slope -0.14, intercept 2.12, R^2 = 0.44. (2) Share with a bachelor's degree: slope -1.44, intercept 1.08, R^2 = 0.37. (3) Rural index: slope 0.83, intercept 0.14, R^2 = 0.60. Each has 3,145 counties. Which variable appears to have the strongest correlation with vote share? Steps: in a simple linear regression, R^2 = r^2, so the largest R^2 means the largest |r|, regardless of the slope sizes (slopes depend on units). Compare 0.44, 0.37, 0.60: the rural index wins. Answer: C.
- Q23. In Philadelphia County, Trump received 19.94% of the total vote (share 0.1994), and the county's share of residents with a bachelor's degree was 34.6% (0.346). Using regression (2), whose fitted line is vote share = 1.08 - 1.44 x bachelor's share, compute the residual for Philadelphia, to two decimals. Steps: first the prediction: 1.08 - 1.44 x 0.346 = 1.08 - 0.498 = 0.58. Then residual = actual - predicted = 0.1994 - 0.58 = -0.38. The negative sign means Philadelphia sits BELOW the line: it voted less for Trump than its education level predicts. Answer: A.
- Q24. Suppose the income variable in regression (1) is redefined as log(average household income in THOUSANDS of dollars) instead of in dollars: a county with $80,000 average income now gets log(80) instead of log(80000). Compute the new estimated slope, to two decimals. Steps: log(income / 1000) = log(income) - log(1000), so the new input is the old input minus a constant (about 6.91). Subtracting a constant from x shifts every x by the same amount; the line's tilt does not change, only the intercept absorbs the shift. The slope stays at its original value. Answer: B (-0.14).
- Q25. Used car pricing: a dataset of 475 cars has price (thousands of dollars), mileage (thousands of miles), and a luxury index (0 to 1). corr(price, mileage) = -0.65 and corr(price, luxury) = 0.45. Which statements are true? (Select all that apply.) (A) The linear correlation between price and mileage is stronger than between price and luxury. (B) The correlation between price and mileage is significantly lower than 0. (C) Mileage explains more variation in prices than the luxury index does. Steps: strength ignores sign: |-0.65| > |0.45|, so (A) is true. With n = 475, an r of -0.65 is enormously far from 0 (its SE is roughly 1/sqrt(475), about 0.046), so (B) is true. Explained variation is r^2: mileage 0.65^2 = 0.42 beats luxury 0.45^2 = 0.20, so (C) is true. Answer: A, B and C.
- Q26. You run two least squares regressions with price as the output: (a) mileage as the input, (b) luxury index as the input. The SD of mileage is 1.95 and the SD of the luxury index is 0.15. Which regression will have the larger estimated slope in absolute magnitude? Steps: the slope formula is b1 = r x (sY / sX), and sY (the SD of price) is the same in both regressions, so compare r / sX. Mileage: 0.65 / 1.95 = 0.33. Luxury: 0.45 / 0.15 = 3.0. Even though luxury's correlation is smaller, its tiny sX makes its slope about nine times bigger in size. Answer: B, regression (b) on luxury.
- Q28. For the car data, you first fit by hand the trial line y-hat = 5.00 - 0.75x (price on mileage) and plot the residuals against x. The residuals RISE with x (negative at low mileage, positive at high mileage), and their mean is zero. You now refit by least squares, getting y-hat = b0 + b1 x. How does b1 compare with -0.75? Steps: negative residuals at low x mean actual prices are below the trial line there (the line is too HIGH); positive residuals at high x mean the line is too LOW there. So the trial line falls too fast: its downslope is too steep. Flattening it means a slope closer to zero, i.e., greater than -0.75. This is the same rule as: least squares residuals must have zero correlation with x, and these residuals still correlate positively with x. Answer: A, b1 > -0.75.

## CHAPTER 4: ESTIMATION, TESTING, PREDICTION

- Q13. In the same Advantage+ output, the standard error of the slope on advantage (--c--) is blanked out. The slope estimate is 0.08612 and its t value is 2.091. What is the SE, to two decimal places? Steps: the t-statistic is built as t = estimate / SE. Rearrange: SE = estimate / t = 0.08612 / 2.091 = 0.0412, which rounds to 0.04. Answer: A.
- Q22. What is the 95% confidence interval for the slope coefficient in regression (3), the rural index? (Slope 0.83, SE 0.35; round to two decimals.) Steps: a confidence interval is estimate +/- multiplier x SE. With 3,145 observations, use the normal multiplier 1.96: 0.83 +/- 1.96 x 0.35 = 0.83 +/- 0.69, giving [0.14, 1.52]. Answer: A.
- Q27. Same two regressions as Q26: price on mileage (a) and price on luxury (b). Which estimated slope has the larger standard error? Steps: SE(b1) = s / sqrt(Sxx), where Sxx is the total squared spread of the x values, which grows with n and with sX^2. Luxury's sX is only 0.15, versus 1.95 for mileage, so its Sxx is far smaller (about 169 times smaller), making its slope's standard error much larger. Less spread in x means the line's tilt is pinned down less precisely. Answer: B, regression (b) on luxury.